Department of Artificial Intelligence

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Concentration Inequalities (AI2200)

Lecture 06

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Sub-Gaussian Random Variables

Theorem 1 (Equivalent Statements for sub-Gaussian Random Variables)

The following statements are equivalent.

1. *There exists a constant* $K_1 > 0$ *such that* $\mathbb{P}\left(\{|X| \geq \varepsilon\}\right) \leq 2\, e^{-\varepsilon^2/K_1^2} \quad \forall\, \varepsilon > 0.$
2. *There exists a constant* $K_2 > 0$ *such that* $\left(\mathbb{E}\left[|X|^p\right]\right)^{1/p} \leq K_2 \sqrt{p} \quad \forall\, p \geq 1.$
3. *There exists a constant* $K_3 > 0$ *such that* $\mathbb{E}\left[e^{t^2 X^2}\right] \leq e^{t^2 K_3^2} \quad \forall\, t \in \left[-\frac{1}{K_3}, \frac{1}{K_3}\right].$
4. *There exists a constant* $K_4 > 0$ *such that* $\mathbb{E}\left[e^{X^2/K_4^2}\right] \leq 2.$
5. **(Applicable if $\mathbb{E}[X] = 0$)**
   There exists a constant $K_5 > 0$ *such that* $\mathbb{E}\left[e^{tX}\right] \leq e^{K_5^2 t^2} \quad \forall\, t \in \mathbb{R}.$

Notes:

- If any of the above statements holds for some constant K, it also holds for a larger constant than K
- If $K_1^\star, \ldots, K_5^\star$ denote the best constants, then for any $i \neq j$, there exist universal constants c_{ij}, C_{ij} such that

$$c_{ij} \leq K_i^\star / K_j^\star \leq C_{ij}.$$

Here, c_{ij} and C_{ij} do not depend on the distribution of X

A Corollary to Theorem 1

- If X is a bounded random variable, i.e., there exists $M > 0$ such that

$$\mathbb{P}\Big(\{|X| \leq M\Big) \; = \; 1,$$

then X is sub-Gaussian

Proof of corollary:

- For any $p \geq 1$, we note that

$$\mathbb{P}\Big(\{|X|^p \leq M^p\}\Big) \; = \; 1, \qquad\qquad \mathbb{E}\Big[|X|^p\Big] \; \leq \; M^p \; \leq \; M^p \, p^{p/2}.$$

- We therefore have

$$\Big(\mathbb{E}\Big[|X|^p\Big]\Big)^{1/p} \; \leq \; M\sqrt{p} \qquad \forall \, p \geq 1.$$

Sub-Gaussian Norm

Definition (Sub-Gaussian Norm)

For a sub-Gaussian random variable X, its **sub-Gaussian norm** is defined as

$$\|X\|_{\psi_2} := \inf\left\{K > 0 : \mathbb{E}\left[e^{X^2/K^2}\right] \leq 2\right\}.$$

- If X is sub-Gaussian, then αX is sub-Gaussian for all $\alpha \in \mathbb{R}$. Furthermore,

$$\|\alpha X\|_{\psi_2} = |\alpha| \, \|X\|_{\psi_2}.$$

- If X, Y are sub-Gaussian with sub-Gaussian norms $K_X = \|X\|_{\psi_2}$ and $K_Y = \|Y\|_{\psi_2}$ respectively, then

$$\|X + Y\|_{\psi_2} \leq \|X\|_{\psi_2} + \|Y\|_{\psi_2} = K_X + K_Y.$$

- More generally, for any $a_1, \ldots, a_n \in \mathbb{R}$ and sub-Gaussian random variables $X_1, \ldots, X_n$,

$$\left\|\sum_{k=1}^{n} a_k X_k\right\|_{\psi_2} \leq \sum_{k=1}^{n} |a_k| \, \|X_k\|_{\psi_2}.$$

Sub-Gaussian Norm

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$$\|X\|_{\psi_2} \; := \; \inf \left\{ K > 0 : \; \mathbb{E}\left[e^{X^2/K^2}\right] \; \leq \; 2 \right\}.$$

- If X, Y are **independent** and sub-Gaussian with $\|X\|_{\psi_2} = K_X$ and $\|Y\|_{\psi_2} = K_Y$ respectively, then

$$\|X+Y\|_{\psi_2} \; \leq \; \sqrt{\|X\|_{\psi_2}^2 + \|Y\|_{\psi_2}^2} \; = \; \sqrt{K_X^2 + K_Y^2}.$$

Proof:

- Without loss of generality, let $\mathbb{E}[X] = 0 = \mathbb{E}[Y]$
- Using Property 5 of Theorem 1, we have that there exists a universal constant $c > 0$ such that

$$\forall\, t \in \mathbb{R}, \qquad \mathbb{E}[e^{tX}] \; \leq \; e^{ct^2 K_X^2}, \qquad \mathbb{E}[e^{tY}] \; \leq \; e^{ct^2 K_Y^2}.$$

- Then, we have

$$\mathbb{E}\left[e^{t(X+Y)}\right] \; \leq \; e^{ct^2 (K_X^2 + K_Y^2)},$$

  from which it follows that

$$\|X+Y\|_{\psi_2}^2 \; \leq \; K_X^2 + K_Y^2.$$

Hoeffding's Inequality

Theorem 2 (Hoeffding's Inequality)

Suppose that $X_1, \ldots, X_n$ are independent and sub-Gaussian with sub-Gaussian norms $K_1, \ldots, K_n$ respectively. Let $\mu_1, \ldots, \mu_n$ denote the means of $X_1, \ldots, X_n$ respectively. Then, for any $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$,

$$\mathbb{P}\left(\left\{\left|\sum_{\ell=1}^{n} \alpha_\ell (X_\ell - \mu_\ell)\right| > \varepsilon\right\}\right) \leq 2\exp\left(-\frac{c\varepsilon^2}{\sum_{\ell=1}^{n} \alpha_\ell^2 (K_\ell^2 + \mu_\ell^2)}\right) \qquad \forall\, \varepsilon > 0,$$

where c is a universal constant that does not depend on the distributions of $X_1, \ldots, X_n$.

Proof of Theorem 2

- We will apply Theorem 1 to the random variable

$$X = \sum_{\ell=1}^{n} \alpha_\ell (X_\ell - \mu_\ell).$$

- For each $\ell \in \{1, \ldots, n\}$, we have

$$\|\alpha_\ell (X_\ell - \mu_\ell)\|_{\psi_2}^2 \leq \alpha_\ell^2 \left(K_\ell^2 + \mu_\ell^2\right).$$

- As a consequence, we have

$$\mathbb{E}\left[\exp\left(t \sum_{\ell=1}^{n} \alpha_\ell (X_\ell - \mu_\ell)\right)\right] = \prod_{\ell=1}^{n} \mathbb{E}[e^{t\alpha_\ell (X_\ell - \mu_\ell)}] \overset{\text{P5, Th.1}}{\leq} \prod_{\ell=1}^{n} e^{t^2 \alpha_\ell^2 \|\alpha_\ell (X_\ell - \mu_\ell)\|_{\psi_2}^2} \leq \prod_{\ell=1}^{n} e^{t^2 \alpha_\ell^2 (K_\ell^2 + \mu_\ell^2)}$$

- Using the equivalence between parts 1, 5 of Theorem 1 for the random variable X, we arrive at the desired result

Application: Multi-Armed Bandits

- Suppose that there are K slot machines, each generating sub-Gaussian rewards
- Let the reward distribution of slot machine k be denoted by ν_k
- At each round $t \in \{1, 2, \ldots\}$, you may play any one of the slot machines
- Let A_t be the slot machine played at time t, and let R_t be the corresponding reward generated
- By playing for a total of T rounds, the goal is to maximise the expected total reward:

$$\max \ \mathbb{E}\left[\sum_{t=1}^{T} R_t\right].$$